Embeddings and Vector Databases

Embeddings

An embedding model maps a piece of text to a fixed length vector of numbers so that texts with similar meaning end up close to each other. The model all-MiniLM-L6-v2 from the Sentence Transformers library produces vectors with 384 dimensions and is small enough to run on a laptop without a graphics card. Its input limit is about 256 word pieces, so longer text is truncated.

Distance metrics

Vector databases rank results by distance. Cosine distance equals one minus the cosine similarity of the two vectors. A distance near zero means the texts are almost identical in meaning, while a distance near one means they are unrelated. Because cosine distance ignores vector length, it is the usual choice for sentence embeddings.

Persistence with Chroma

Chroma is an open source vector database that can run embedded inside a Python process. A persistent client stores the collection on disk in a directory, so the index survives restarts and ingestion only needs to run again when documents change. Each record holds an id, the chunk text, its embedding and a metadata dictionary. Using a deterministic id built from file name, page and chunk index lets re-ingestion update records in place instead of creating duplicates. Records of a file should be deleted before the file is re-indexed, otherwise chunks from an older version can remain in the collection.